HOTEL BOOKINGS DATA ANALYSIS

From Raw Data to Business Insight

Subject: Data Analytics Visulization

Event: InsightVista

Group: DataVerse

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Data Observations

47,883
booking records

29
columns / variables

2
hotels: City & Resort

38%
bookings cancelled

- Period covered: arrivals in 2015, 2016 and 2017; peak months are August and July
- Resort Hotel holds 71% of bookings (33,955) vs City Hotel 29% (13,928)
- Online travel agents are the biggest channel: 26,948 bookings, 56% of the data
- Average daily rate (adr) is about 100, but the maximum is 5,400, a clear outlier
- Guests are mostly adults (90,105) with 7,126 children and 637 babies; Portugal, UK and Spain lead by country
- Mix of types: numeric (lead time, nights, adr), categorical (hotel, meal, segment) and flags (is_canceled)

Raw Data to Clean Data: Steps

1 Load & inspect

Checked shape, data types, null counts and summary statistics

2 Handle missing values

Filled or removed blanks in children, country and similar fields

3 Drop unneeded columns

Removed columns with heavy gaps or little analytic value (agent, company, status date)

4 Fix names & types

Renamed lead_time to days_booked_in_advance; set integer and float types

5 Remove invalid rows

Dealt with impossible records such as zero-guest bookings

6 Validate

Re-checked nulls and value ranges, then exported the clean CSV

Solutions & Final Clean Dataset

What we did

- Removed null values, so every field is now complete
- Gave columns clear, consistent names
- Kept only the 29 columns needed for analysis
- Set correct data types for dates, counts and rates
- Flagged outliers (adr up to 5,400) for review in analysis

Final dataset

Check	Result
Rows	47,883
Columns	29
Missing values	0
Hotels	Resort 33,955 / City 13,928
Years	2015 to 2017
File	cleaned_hotel_bookings_final.csv

Dashboard, Graphs & Insights



Power BI dashboard (June filter)

Cancellations climb with longer lead time: cancelled bookings were made 98 days ahead on average vs 73 for kept ones.

Key insights

- Adults make up 91.5% of guests; children 7.8%, babies 0.6%
- Resort Hotel generates 63.6% of hotel nights
- Online TA drives the most cancellations, over 1,000 in the view
- Resort Hotel has far longer booking lead time and waiting list days
- Overall, 38% of bookings cancel; City Hotel 73%, Resort 23%

Real-World Implications & Use Cases

Cut cancellations

Online TA bookings cancel 51% of the time; Non-Refund deposits cancel 95%. Use deposit or flexible-rate rules for risky segments.

Smarter overbooking

Predict cancellations from lead time, segment and hotel to overbook safely and protect revenue.

Pricing & revenue

Track adr by month and season; raise rates in July and August peaks, run offers in low months.

Channel strategy

Shift volume to Direct (19% cancel) and Corporate (18%) to lower dependence on Online TA.

Use case: a hotel revenue manager can use this dashboard to plan staffing, rates and overbooking each month.